

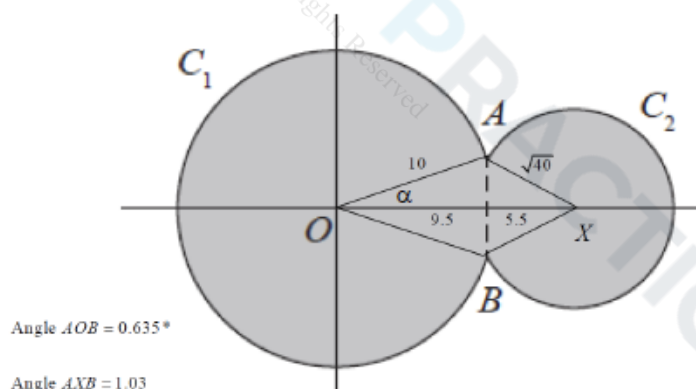
Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	Uses $s = r\theta \Rightarrow 3 = r \times 0.4$	M1	1.2
	$\Rightarrow OD = 7.5 \text{ cm}$	A1	1.1b
		(2)	
(b)	Uses angle $AOB = (\pi - 0.4)$ or uses radius is $(12 - '7.5')$ cm	M1	3.1a
	Uses area of sector $= \frac{1}{2}r^2\theta = \frac{1}{2} \times (12 - 7.5)^2 \times (\pi - 0.4)$	M1	1.1b
	$= 27.8 \text{ cm}^2$	A1ft	1.1b
		(3)	
(5 marks)			
Notes:			
(a)			
M1: Attempts to use the correct formula $s = r\theta$ with $s = 3$ and $\theta = 0.4$			
A1: $OD = 7.5 \text{ cm}$ (An answer of 7.5cm implies the use of a correct formula and scores both marks)			
(b)			
M1: $AOB = \pi - 0.4$ may be implied by the use of $AOB = \text{awrt } 2.74$ or uses radius is $(12 - \text{their '7.5'})$			
M1: Follow through on their radius $(12 - \text{their } OD)$ and their angle			
A1ft: Allow awrt 27.8 cm^2 . (Answer 27.75862562). Follow through on their $(12 - \text{their '7.5'})$			
Note: Do not follow through on a radius that is negative.			

Q2.

Question	Scheme	Marks	AOs
(a)	Solves $x^2 + y^2 = 100$ and $(x-15)^2 + y^2 = 40$ simultaneously to find x or y E.g. $(x-15)^2 + 100 - x^2 = 40 \Rightarrow x = \dots$	M1	3.1a
	Either $\Rightarrow -30x + 325 = 40 \Rightarrow x = 9.5$ Or $y = \frac{\sqrt{39}}{2} = \text{awrt } \pm 3.12$	A1	1.1b
	Attempts to find the angle AOB in circle C_1 Eg Attempts $\cos \alpha = \frac{9.5}{10}$ to find α then $\times 2$	M1	3.1a
	Angle $AOB = 2 \times \arccos\left(\frac{9.5}{10}\right) = 0.635 \text{ rads (3sf)}$ *	A1*	2.1
		(4)	
(b)	Attempts $10 \times (2\pi - 0.635) = 56.48$	M1	1.1b
	Attempts to find angle AXB or AXO in circle C_2 (see diagram) E.g. $\cos \beta = \frac{15 - 9.5}{\sqrt{40}} \Rightarrow \beta = \dots$ (Note $AXB = 1.03$ rads)	M1	3.1a
	Attempts $10 \times (2\pi - 0.635) + \sqrt{40} \times (2\pi - 2\beta)$	dM1	2.1
	$= 89.7$	A1	1.1b
		(4)	
(8 marks)			
Notes:			



(a)

M1: For the key step in an attempt to find either coordinate for where the two circles meet.

Look for an attempt to set up an equation in a single variable leading to a value for x or y .

A1: $x = 9.5$ (or $y = \frac{\sqrt{39}}{2} = \text{awrt } \pm 3.12$)

M1: Uses the radius of the circle and correct trigonometry in an attempt to find angle AOB in circle C_1

E.g. Attempts $\cos \alpha = \frac{9.5}{10}$ to find α then $\times 2$

Alternatives include $\tan \alpha = \frac{\sqrt{100 - 9.5^2}}{9.5} = (0.3286...)$ to find α then $\times 2$

$$\text{And } \cos AOB = \frac{10^2 + 10^2 - (\sqrt{39})^2}{2 \times 10 \times 10} = \frac{161}{200}$$

A1*: Correct and careful work in proceeding to the given answer. Condone an answer with greater accuracy.

Condone a solution where the intermediate value has been truncated, provided the trig equation is correct.

$$\text{E.g. } \sin \alpha = \frac{\sqrt{39}}{20} \Rightarrow \alpha = 0.317 \Rightarrow AOB = 2\alpha = 0.635$$

Condone a solution written down from awrt 36.4° (without the need to shown any calculation.)

E

(b)

M1: Attempts to use the formula $s = r\theta$ with $r=10$ and $\theta = 2\pi - 0.635$

The formula may be embedded. You may see $\frac{2\pi \times 10 + 2\pi \sqrt{40} - 10 \times 0.635}{...}$ which is fine for this M1

M1: Attempts to use a correct method in order to find angle AXB or AXO in circle C_2

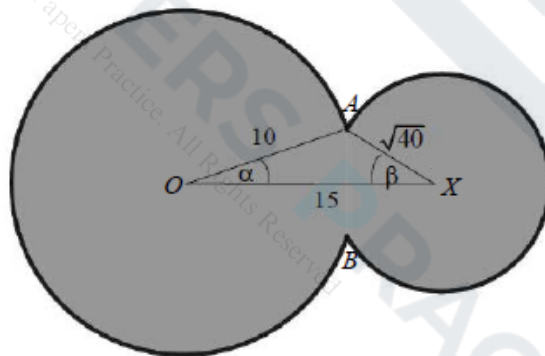
$$\text{Amongst many other methods are } \tan \beta = \frac{3.12}{15 - 9.5} \text{ and } \cos AXB = \frac{40 + 40 - (\sqrt{39})^2}{2 \times \sqrt{40} \times \sqrt{40}} = \frac{41}{80}$$

Note that many candidates believe this to be 0.635. This scores M0 dM0 A0

dM1: A full and complete attempt to find the perimeter of the region.

It is dependent upon having scored both M's.

A1: awrt 89.7



(a)

M1: For the key step in attempting to find all lengths in triangle OAX , condoning slips

A1: All three lengths correct

M1: Attempts cosine rule to find α then $\times 2$

A1*: Correct and careful work in proceeding to the given answer

Q3.

Question Number	Scheme	Marks	AO's
	Attempts any one of $(\pm \overrightarrow{PQ} =) \pm (\mathbf{q} - \mathbf{p}), (\pm \overrightarrow{PR} =) \pm (\mathbf{r} - \mathbf{p}), (\pm \overrightarrow{QR} =) \pm (\mathbf{r} - \mathbf{q})$ Or e.g. $(\pm \overrightarrow{PQ} =) \pm (\overrightarrow{OQ} - \overrightarrow{OP}), (\pm \overrightarrow{PR} =) \pm (\overrightarrow{OR} - \overrightarrow{OP}), (\pm \overrightarrow{QR} =) \pm (\overrightarrow{OR} - \overrightarrow{OQ})$	M1	1.1b
	Attempts e.g. $\mathbf{r} - \mathbf{q} = 2(\mathbf{q} - \mathbf{p})$ $\mathbf{r} - \mathbf{p} = 3(\mathbf{q} - \mathbf{p})$ $\frac{2}{3}(\mathbf{q} - \mathbf{p}) = \frac{1}{3}(\mathbf{r} - \mathbf{q})$ $\mathbf{q} = \mathbf{p} + \frac{1}{3}(\mathbf{r} - \mathbf{p})$ $\mathbf{q} = \mathbf{r} + \frac{2}{3}(\mathbf{p} - \mathbf{r})$	dM1	3.1a
	E.g. $\Rightarrow \mathbf{r} - \mathbf{q} = 2\mathbf{q} - 2\mathbf{p} \Rightarrow 2\mathbf{p} + \mathbf{r} = 3\mathbf{q} \Rightarrow \mathbf{q} = \frac{1}{3}(\mathbf{r} + 2\mathbf{p})^*$	A1*	2.1
		(3)	
(3 marks)			

Notes:

M1: Attempts any of the relevant vectors by subtracting either way around. This may be implied by sight of any one of $\pm(\mathbf{q} - \mathbf{p}), \pm(\mathbf{r} - \mathbf{p}), \pm(\mathbf{r} - \mathbf{q})$ ignoring how they are labelled

dM1: Uses the given information and writes it correctly in vector form that if rearranged would give the printed answer

A1*: Fully correct work leading to the given answer. Allow $\mathbf{OQ} = \dots$ as long as $\mathbf{OQ}$ has been defined as $\mathbf{q}$ earlier.

In the working allow use of $\mathbf{P}$ instead of $\mathbf{p}$ and $\mathbf{Q}$ instead of $\mathbf{q}$ as long as the intention is clear.

Q4.

Question	Scheme	Marks	AOs
	Attempts $\vec{AC} = \vec{AB} + \vec{BC} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k} + \mathbf{i} - 9\mathbf{j} + 3\mathbf{k} = 3\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}$	M1	3.1a
	Attempts to find any one length using 3-d Pythagoras	M1	2.1
	Finds all of $ AB = \sqrt{14}$, $ AC = \sqrt{61}$, $ BC = \sqrt{91}$	A1ft	1.1b
	$\cos BAC = \frac{14 + 61 - 91}{2\sqrt{14}\sqrt{61}}$	M1	2.1
	angle $BAC = 105.9^\circ$ *	A1*	1.1b
		(5)	
(5 marks)			
Notes:			
M1: Attempts to find $\vec{AC}$ by using $\vec{AC} = \vec{AB} + \vec{BC}$ M1: Attempts to find any one length by use of Pythagoras' Theorem A1ft: Finds all three lengths in the triangle. Follow through on their AC M1: Attempts to find BAC using $\cos BAC = \frac{ AB ^2 + AC ^2 - BC ^2}{2 AB AC }$ Allow this to be scored for other methods such as $\cos BAC = \frac{\vec{AB} \cdot \vec{AC}}{ AB AC }$ A1*: This is a show that and all aspects must be correct. Angle $BAC = 105.9^\circ$			

Q5.

Question	Scheme	Marks	AOs
(a)	$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = -3\mathbf{i} - 4\mathbf{j} - 5\mathbf{k} + \mathbf{i} + \mathbf{j} + 4\mathbf{k} = \dots$	M1	1.1b
	$= -2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$	A1	1.1b
		(2)	
(b)	At least 2 of $(AC^2) = 2^2 + 3^2 + 1^2, (AB^2) = 3^2 + 4^2 + 5^2, (BC^2) = 1^2 + 1^2 + 4^2$	M1	1.1b
	$2^2 + 3^2 + 1^2 = 3^2 + 4^2 + 5^2 + 1^2 + 1^2 + 4^2 - 2\sqrt{3^2 + 4^2 + 5^2}\sqrt{1^2 + 1^2 + 4^2} \cos ABC$	M1	3.1a
	$14 = 50 + 18 - 2\sqrt{50}\sqrt{18} \cos ABC$ $\Rightarrow \cos ABC = \frac{50 + 18 - 14}{2\sqrt{50}\sqrt{18}} = \frac{9}{10}^*$	A1*	2.1
		(3)	
	(b) Alternative		
	$AB^2 = 3^2 + 4^2 + 5^2, BC^2 = 1^2 + 1^2 + 4^2$	M1	1.1b
	$\overrightarrow{BA} \cdot \overrightarrow{BC} = (3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}) \cdot (\mathbf{i} + \mathbf{j} + 4\mathbf{k}) = 27 = \sqrt{3^2 + 4^2 + 5^2}\sqrt{1^2 + 1^2 + 4^2} \cos ABC$	M1	3.1a
	$27 = \sqrt{50}\sqrt{18} \cos ABC \Rightarrow \cos ABC = \frac{27}{\sqrt{50}\sqrt{18}} = \frac{9}{10}^*$	A1*	2.1
(5 marks)			
Notes			

(a)

M1: Attempts $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

There must be attempt to add not subtract.

If no method shown it may be implied by two correct components

A1: Correct vector. Allow $-2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ and $\begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix}$ but not $\begin{pmatrix} -2\mathbf{i} \\ -3\mathbf{j} \\ -1\mathbf{k} \end{pmatrix}$

(b)

M1: Attempts to "square and add" for at least 2 of the 3 sides. Follow through on their $\overrightarrow{AC}$

Look for an attempt at either $a^2 + b^2 + c^2$ or $\sqrt{a^2 + b^2 + c^2}$

M1: A correct attempt to apply a correct cosine rule to the given problem; Condone slips on the lengths of the sides but the sides must be in the correct position to find angle ABC

A1*: Correct completion with sufficient intermediate work to establish the printed result.

Condone different labelling, e.g. $ABC \leftrightarrow \theta$ as long as it is clear what is meant

It is OK to move from a correct cosine rule $14 = 50 + 18 - 2\sqrt{50}\sqrt{18} \cos ABC$

via $\cos ABC = \frac{54}{2\sqrt{50}\sqrt{18}}$ o.e. such as $\cos ABC = \frac{(5\sqrt{2})^2 + (3\sqrt{2})^2 - (\sqrt{14})^2}{2 \times 5\sqrt{2} \times 3\sqrt{2}}$ to $\cos ABC = \frac{9}{10}$

Alternative:

M1: Correct application of Pythagoras for sides AB and BC or their squares

M1: Recognises the requirement for and applies the scalar product

A1*: Correct completion with sufficient intermediate work to establish the printed result